THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A person with myopia has a far point at 100 cm. The lens needed has a power of 1

(a) +10 D **(b)** +1 D **(c)** -0.1 D **(d)** -1 D

2. A beam of white light is passed through a prism and a screen shows 7 colours. Replacing white by red light gives 1

(a) white light **(b)** a single red patch

(c) no light **(d)** 7 colours

3. If the atmosphere of the earth were absent, the day sky would appear 1

(a) blue **(b)** dark **(c)** red **(d)** white

4. A person's near point is 50 cm. To read at 25 cm, he needs a lens of power 1

(a) +0.5 D **(b)** +2 D **(c)** +4 D **(d)** -2 D

5. The deviation of violet light in a prism is greater than that of red because 1

(a) red is absorbed

(b) violet has lower frequency

(c) violet travels faster in glass

(d) the refractive index for violet is higher

6. The Sun at noon appears white because 1

(a) no scattering occurs

(b) light travels a short path and little blue scatters

(c) violet is absorbed

(d) red gets scattered

7. A lens of power -2.5 D has focal length 1

(a) -40 cm **(b)** +40 cm **(c)** -2.5 cm **(d)** -25 cm

8. In glass, the colour of light with the highest speed is 1

(a) red **(b)** green **(c)** blue **(d)** violet

9. Fine particles in the atmosphere scatter mainly 1

(a) blue light **(b)** red light **(c)** yellow light **(d)** no light

10. A myopic person's eyeball is too long. His eye forms the image of a distant object 1

(a) behind the retina **(b)** on the retina

(c) in front of the retina **(d)** at infinity

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A person has a near point at 40 cm. Find the focal length of the lens needed to read at 25 cm. 2

12. Why does a person with hypermetropia hold a book farther away to read? 2

13. Why is the sky above a mountain-top sometimes darker blue? 2

14. A person can see clearly only up to 2 m. Find the power of the lens to correct this. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A person cannot see beyond 1.5 m clearly. Find the focal length, power and nature of the corrective lens. 3

16. Explain why the colour of the Sun is red during sunset but white at noon. 3

17. What is the minimum distance at which objects can be seen clearly? Why do we hold things farther than a few cm? 3

OR

A hypermetropic person has a near point at 75 cm. Find the power of the lens to read at 25 cm. 3

18. Explain how a prism in the inverted position recombines the spectrum. 3

OR

Give two reasons why the planets can be seen without twinkling. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A person has a near point of 1 m and a far point at infinity. 5
(a) Name the defect
(b) find the lens power to read at 25 cm
(c) state the nature of the lens.

OR

A person's far point is at 80 cm and near point at 25 cm. Find the power of the lens needed to see distant objects. Is any lens needed for reading? 5

20. A short-sighted person has a far point at 40 cm. 5
(a) Name the defect
(b) find the power of the lens to see far objects
(c) explain why the lens is concave.

OR

Explain the terms: dispersion, scattering, refraction and atmospheric refraction with one example each. 5

END OF QUESTION PAPER — Total: 40 Marks